Finding and Verifying Inverse Functions

Finding inverses of linear, rational, radical and power rules, then checking them by composition

- To find an inverse rule, write $x = f(y)$ and then solve for y; what you get is $f^{-1}(x)$.
- Leave every answer as an exact expression — no decimals, no rounding.
- Write the algebra that produced the answer, not just the answer.

1. Find the inverse rule for $f(x) = 5x - 8$.

Answer: ________________

2. Find the inverse rule for $g(x) = \dfrac{x}{4} + 3$.

Answer: ________________

3. Let $p(x) = 2x + 9$ and $q(x) = \dfrac{x-9}{2}$.

(a) Simplify $p(q(x))$ completely.

Answer: ____________________

(b) Explain what the part (a) result tells you about how p and q are related.

Answer: ____________________

4. Find the inverse rule for $r(x) = \dfrac{2x+1}{x-3}$.

Answer: ____________________

5. Find the inverse rule for $s(x) = \sqrt{x+7} - 2$, whose inputs satisfy $x \geq -7$.

Answer: ____________________

6. For $m(x) = 4x + 3$, find $m^{-1}(19)$.

Answer: ________________

7. Find the inverse rule for $t(x) = \dfrac{3 - 2x}{5}$.

Answer: ________________

8. Find the inverse rule for $u(x) = \dfrac{4}{x + 6} + 1$.

Answer: ________________

9. Let $w(x) = (x-4)^3 + 2$. A classmate claims that $w^{-1}(x) = \sqrt[3]{x-2} + 4$. Test the claim by simplifying $w\big(w^{-1}(x)\big)$, and write the simplified result.

Answer: ____________________

10. Let $n(x) = 7 - 3x$.

(a) Find $n^{-1}(x)$.

Answer: ____________________

(b) Simplify $n\big(n^{-1}(x)\big)$ to confirm your rule.

Answer: ____________________

11. Let $v(x) = x^2 - 6x + 5$, with inputs restricted to $x \geq 3$.
(a) Confirm the claim $v^{-1}(x) = 3 + \sqrt{x+4}$ by simplifying $v\big(v^{-1}(x)\big)$.

Answer: __________________

(b) Explain why the restriction $x \geq 3$ has to be stated for v to have an inverse at all.

Answer: __________________

12. Let $f(x) = \dfrac{x-3}{2x+1}$.
(a) Find $f^{-1}(x)$.

Answer: __________________

(b) One input value has to be excluded from the domain of f^{-1}. Name it, and explain what goes wrong there.

Answer: __________________